

Exp No: 5 ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;
void setup() {
  //configure pin2 as an input and enable the internal pull-up resistor
  pinMode(12, INPUT_PULLUP);
  pinMode(1,OUTPUT);
  pinMode(2,OUTPUT);
  pinMode(3,OUTPUT);
  pinMode(4,OUTPUT);
  pinMode(5,OUTPUT);
  pinMode(6,OUTPUT);
  pinMode(7,OUTPUT);
}
void loop() {
  //read the pushbutton value into a variable
  int sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x0=true;
  }
  while(x0){
    zero();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x1=true;
      x0=false;
    }
  }
}
```

```
while(x1){
one();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x2=true;
x1=false;
}
}
while(x2){
two();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x3=true;
x2=false;
}
}
while(x3){
three();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x4=true;
x3=false;
}
}
while(x4){
four();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x5=true;
x4=false;
```

```
}  
}  
while(x5){  
    five();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x6=true;  
        x5=false;  
    }  
}  
while(x6){  
    six();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x7=true;  
        x6=false;  
    }  
}  
while(x7){  
    seven();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x8=true;  
        x7=false;  
    }  
}  
while(x8){  
    eight();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {
```

```
x9=true;
x8=false;
}
}
while(x9){
  nine();
  sensorVal = digitalRead(12);
  if (sensorVal == LOW) {
    x0=true;
    x9=false;
  }
}

}
void zero()
{
  for(int i=1;i<7;i++)
  {
    digitalWrite(i,HIGH);
    digitalWrite(7,LOW);
  }
  delay(delay_time);
}
void one()
{
  digitalWrite(1,LOW);
  digitalWrite(2,HIGH);
  digitalWrite(3,HIGH);
  digitalWrite(4,LOW);
  digitalWrite(5,LOW);
```

```
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}
void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}
void three()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}
void four()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
```

```
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void five()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void six()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void seven()
```

```
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
}  
  
void eight()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}  
  
void nine()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);
```

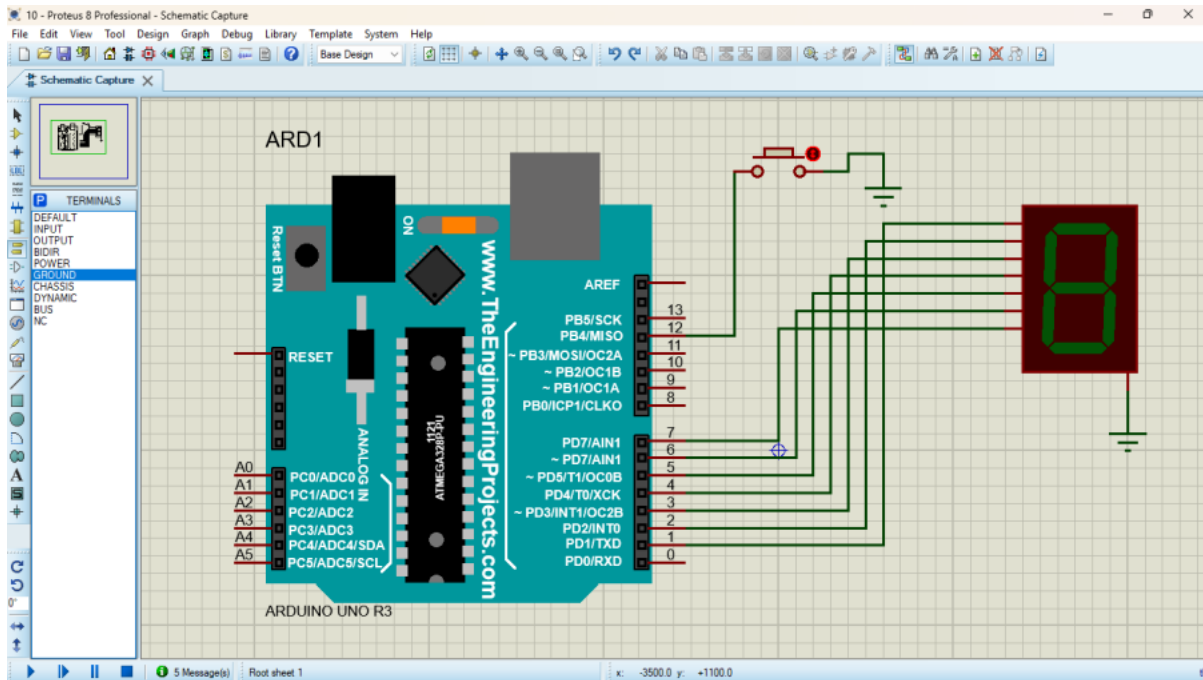
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delay(delay_time);

}

```

SIMULATION:



OUTPUT:

